

Bias FAQ



Duncanamps

Site: **Duncanamps**

Saved  Today at 21:38

...

HOW TO BIAS YOUR AMPLIFIER

DISCLAIMER: LORD VALVE (W. WHITTAKER DBA NBS ELECTRONICS) WILL TAKE NO RESPONSIBILITY FOR ANY INJURIES OR DAMAGE SUFFERED BY ANYONE AS A RESULT OF THE MISINTERPRETATION OR MISAPPLICATION OF THE INFORMATION CONTAINED IN THIS ARTICLE. PROPER HIGH-VOLTAGE SAFETY PROCEDURES AND PRECAUTIONS MUST BE PRACTICED AT ALL TIMES WHEN SERVICING VACUUM TUBE AMPLIFIERS!! IF YOU ARE NOT *COMPLETELY* FAMILIAR WITH THESE PRECAUTIONS, SEEK GUIDANCE FROM SOMEONE WHO IS!

THE CATHODE RESISTOR METHOD

This is the method that is best for hobby techs and do-it-yourselfers. It is far and away the safest of the two methods, and can be successfully done with medium- and even low-quality test equipment. It is performed by reading the cathode current through each power tube. The cathode current is composed of the plate current *plus* the screen current. Plate current can be identical on two tubes (tubes are matched by plate current readings) while one tube is drawing more screen current; with this method, the readings will appear to indicate a mismatch when such is not actually the case. Since the cathode current will always be higher than the actual plate current, the readings obtained with this method will tend to make you set the tubes a little colder than your calculations will indicate that they are. This promotes slightly more conservative operation, which is beneficial to tube life. If you feel that the inaccuracy is significant, or if you'd just like to know how large it is for your particular amp, you can always read the voltage drop across each screen resistor, figure the screen current ($I = E/R$) and then subtract it from the cathode reading. Note that these instructions assume that your amplifier is biased by applying a negative voltage to the control grids; cathode-biased amplifiers cannot be adjusted other than by changing the value of the cathode resistor(s) so this method does not apply to them. BE AWARE THAT THE ACCURACY OF THE RESULTS YOU OBTAIN WITH *ANY* BIASING METHOD WILL BE DIRECTLY AFFECTED BY THE QUALITY OF YOUR TEST EQUIPMENT, AND YOUR SKILL IN USING IT. If any part of the following instructions doesn't make sense to you, seek help from someone with more experience. Tech support is available from NBS Electronics AFTER 1:30 PM Denver (Mountain) time, Monday through Saturday. The number is 303-778-1156; ask for Lord Valve.

WHAT YOU WILL DO:

A) Replace the ground wire on each power tube socket with a 1-ohm resistor. B) Read the voltage drop across this resistor (in millivolts) with your DMM. C) Read the plate voltage. D) Use the above readings to calculate the static dissipation wattage. E) Adjust the bias to obtain the best tone, while keeping the tubes within specifications.

A SUGGESTION: You may want to practice taking these readings and making these adjustments with your old tubes still in the amp, or with a spare (used) set. That way, you won't fry your new tubes if you make a mistake.

On some sockets, the pins are numbered on the bottom (terminal) side; it is sometimes difficult to tell which pins the numbers go with. The best way to tell which pin you are looking at is to count clockwise from the notch on the locator "keyhole" in the center of the socket, with the first terminal clockwise from the notch being pin ONE. This assumes that you are looking at the sockets from the BOTTOM, or UNDERSIDE.

Most guitar amplifiers use output tubes which have the same (or very similar) basing. ("Basing" refers to the order in which the internal elements of the tube are connected to the pins on the bottom of the tube.) The 6L6, 7581A, 6V6, 6550, EL34, 5881, KT66, KT88, KT90, KT100, etc. are common guitar amplifier power tubes, and are all easily biased with this method. If you have an amp which uses power tubes which are not listed above, you will need to consult a spec manual (the RCA RC-30 Receiving Tube Manual is one of the best) for the basing, and adjust these instructions accordingly.

You'll need a 1-ohm resistor for each power tube in the amp. All of the tubes listed above have their cathodes on pin EIGHT, which will be grounded to the chassis. On some amps, such as Marshalls, pin ONE will be tied to pin EIGHT, and both will be grounded. On older Fenders, pin ONE is usually used as a tie-point for the 1.5K grid-stopper resistor, and the negative bias voltage will be on this pin. **DO NOT GROUND PIN ONE ON A FENDER AMPLIFIER**, or you'll get a big surprise. (Expensive, too. ;-)

REMOVE the ground wire from pin EIGHT on each tube, and REPLACE it with a 1-ohm resistor. On older Fenders, the ground wire is a piece of copper braid; unsolder it from the socket pin but **don't** cut it off where it attaches to the chassis. Solder the resistor to pin EIGHT, and attach the free end of the resistor to the ground braid you unsoldered from pin EIGHT. Repeat this for all of the power tube sockets. I prefer to use 2-watt resistors because they have thicker leads which will take more abuse, but half-watters will work just fine. The accuracy of your measurements will be directly related to the tolerance of these resistors; precision 1% (or better) types are suggested.

A WORD ABOUT TAKING READINGS: It is vital that your probe tips make **good** contact with the pins you're attempting to read. Tube-socket terminals often have a residual coating of non-conductive flux on them, and it is necessary to push the probe against the terminal hard enough to break through this coating. Most test probes supplied with today's meters are fairly blunt; if you can come up with a set of "insulation piercing" test probes, these will help solve this problem. Don't go overboard when pressing the probes against terminals, however...the probe tip may suddenly slip off the terminal and slide down against the chassis while the side of the metal prong is still touching the terminal. This will result in a dead short from that terminal to ground, and if you're reading plate or screen voltage the resulting spark (and loud popping noise) might make you jerk back reflexively, pulling the chassis off your workbench and into your lap, injuring you or breaking your tubes. **BE CAREFUL!** Also be aware that amp chassis surfaces are usually dirty or corroded, so the advice above goes double for touching a probe to the chassis (ground). You may even want to take a small flat file and scrape a nice shiny spot on the chassis in a convenient place, to aid you in making a good ground contact.

Turn your amp on, but leave it on STANDBY. Set your DMM to the highest DCV scale, ground the black probe to the chassis, and take a reading from pin FIVE of any power tube socket. You should see a negative voltage in the -35 to -50 volt range if the amp has EL34s, or in the -45 to -60 volt range if the amp uses 5881s, 6L6s, or KT66s. KT88s, 6550s, KT90s, and KT100s can have bias ranges that go as high as -100 volts. Amps which use 6V6s will usually have bias supplies which produce voltages that are similar to EL34 amps...but not always. Note that you should **not** have any power tubes installed in your amp yet.

First, locate the bias trimmer. (Possibly a little square blue thingy with a screwdriver-adjust slot in the center, or a round black thing that stands on three legs, or, for an old Fender, a full-sized pot with a screwdriver-adjust slot on both sides; newer PCB-type Fenders use three-leg horizontal trimpots, if they have a bias-adjust pot at all.) Next, adjust the bias control until you have MAX NEGATIVE voltage on pin FIVE. (In other words, rotate the bias trimmer until you obtain the highest negative voltage that the bias supply is capable of delivering.) Note that on some really old amps, the bias supply may be controlled by the standby switch; if you see **no** negative voltage on pin FIVE, you may have one of these amps. This is a poor design; see a tech about having the bias supply moved to the "hot" side of the standby switch. Install your tubes (the amp is still on STANDBY, remember) and wait a few minutes for them to warm up. Take the amp off STANDBY and make sure your DMM is still set to the highest DCV scale; take a reading between the chassis (ground) and pin THREE on any power tube socket. Remember, the BLACK probe always goes on the CHASSIS. Write this voltage down; you'll need it later.

Now, set your DMM to the lowest DCV scale (usually 200 mV) and take a reading across the 1-ohm resistor(s). (This reading can be interpreted directly in milliamperes, because one millivolt across one ohm equals one milliamp. Ohm's law says so, and you ain't gonna argue with **that**, are ya? ;-) It'll be pretty low, because you have the bias trimmer set to max neg voltage.

Adjust the bias trimmer ("pot") until you get a reading across the 1-ohm resistor(s) somewhere in the 30-40 mV range, for everything but 6V6s or EL-84s. For 6V6s, you'll want to start out at around 20 mA and work upward from there. Note that the polarity of this reading is unimportant; only the numerical value means anything. (If you put the black probe on the side of the resistor that is grounded to the chassis, you will get a POSITIVE reading.)

MULTIPLY the voltage you read on pin THREE earlier by the reading you just obtained from the 1-ohm resistor. (Example: 450 Volts times 35 milliamps, or .035 Amperes.) This will give you the STATIC DISSIPATION WATTAGE at which the tube is operating. (It'll be wrong, but more on that later.) The above example gives a static dissipation of 15.75 WATTS, which is well within specs for an EL34 (fairly cold, in fact) or a 5881/6L6. See TABLE "A" (at the end of this article) for suggested static dissipation wattages for most of the common octal-based tubes discussed here. To sum up what this calculation is, PLATE VOLTAGE times CATHODE CURRENT equals STATIC DISSIPATION (IDLING) WATTAGE. It is important not to exceed the tube manufacturer's specification for this parameter, because tube life will be shortened. At extreme settings, tube life will be measured in MINUTES...be advised.

Take another reading from pin THREE (remember to set your meter on the HIGHEST DCV scale before you do!) and write it down. This new reading should be LOWER than the first reading you took, because the tubes are drawing more current now and the plate voltage will drop somewhat. Multiply this new reading by the value you measured across the 1-ohm resistor(s); this will give you the idling (static) wattage. The cooler you run the tubes, the longer they'll last. If you dig the way the amp sounds when the tubes are idling at only 12 watts, fine...don't worry about it. (6V6s, though, will be running fairly *hot* at 12 watts.)

Remember, each time you adjust the bias control, you'll have to take a new reading from BOTH the 1-ohm resistor *and* the plate (pin THREE) and multiply them to see how hot the tubes are running. You can play your guitar through the amp each time you adjust the bias, and see how you like it. You can even adjust the bias by ear, and then take readings as outlined above to see if the tubes are being operated within their ratings. If you find that you only like the tone when the tubes are operating near their limits, you may decide to trade some tube lifetime for the tone you seek. If you like the tone with the tubes running cold, you'll obtain significant extra tube life that way. It's *your* call.

If you see a few milliamps difference between the readings on the 1-ohm resistors, don't sweat it; this could be due to poor matching (not a factor if you bought 'em from *me* :), differences in screen current between the tubes, or differing leg impedances in the output tranny's primary. (All of those things are fairly common in guitar amps.) Note that for an amplifier which uses four (or more) power tubes, balance between the two sides is more important than having identical readings from socket to socket. You should add the readings for each pair; if the left pair is close to the right pair, things are fine. If the left pair reads 32 and 34 milliamps (total = 66) and the right pair reads 35 and 31 milliamps (total = 66) then you've got a nicely balanced output stage, even though some of the tubes are running slightly hotter or colder than others. Having the currents balanced on the two legs of the tranny helps eliminate 120 Hz power-supply ripple from the output. Note that you can swap the tubes around to obtain the best current balance, since you can take individual readings on each socket. If you see a large difference between them (say, 8-12 milliamps) this means you need to find out why this difference exists. One thing you can do is SWAP the tubes into the opposite sockets and take new readings. If the bogus readings are consistent on the SOCKETS, then you'll need to look at the circuitry to find out the cause. If the readings MOVE with the TUBES, you can be fairly sure you have a poorly-matched pair/quad.

Once you have everything adjusted to your taste and you're sure the tubes are being operated within specifications, leave the amp fully powered up for three or four hours. Eyeball the tubes every fifteen minutes or so, to make sure the plates aren't turning red. You are doing this to let the tubes "settle" into their new operating conditions; at the end of the settling period, take a final set of readings to make sure everything is still OK. If any readings have drifted significantly, readjust the bias accordingly. Note that the incoming line voltage directly affects all of the voltages in the amp; you may want to read the line voltage occasionally to see if this is happening. Line voltage will drop a bit around supper time (lots of juice being used for cooking) and also after sunset. If the line was 120VAC when you completed your biasing procedure and it's 117VAC when you take your final readings after the settling period, expect to see a corresponding small drop in your measurements.

You may decide to purchase a "bias-probe" type device; this is a gizmo which consists of an "interruptor" socket/plug assembly that goes between the tube(s) and the amp's socket(s). This test adaptor will have a couple of testleads hanging out through a hole in the side, for connection to your meter(s). If you do get one of these, there is no need to install the 1-ohm resistors on the tube sockets as outlined above. You can use the readings obtained from the adaptor sockets in place of the readings normally taken across the one-ohm resistors. BE AWARE THAT THERE ARE TWO TYPES OF THESE ADAPTORS COMMONLY AVAILABLE. One type **breaks** the cathode connection, and instructs you to connect the testleads to the CURRENT jacks on your meter. The other type contains a one-ohm resistor in series with the cathode pins, with the testleads connected to either side of the resistor; this type instructs you to connect the testleads to the VOLTAGE jacks on your meter. It has been my experience that some amps (especially old Marshalls) do not react well to having several feet of wire inserted in series with the power tube cathodes, and will oscillate like crazy. Therefore, if you decide to get a set of these test adaptors, get the ones which use an internal 1-ohm resistor.

REMEMBER...THERE ARE VOLTAGES PRESENT INSIDE EVEN THE SMALLEST TUBE AMPLIFIER WHICH WILL KILL YOUR ASS JUST AS DEAD AS A HAND GRENADE WILL!! If you're not familiar with high-voltage safety, seek guidance from someone who is. BTW, an oven mitt or a pot-holder (real men like me use welding gloves) will come in handy for handling hot power tubes if you need to switch sockets; you don't want to let the tubes cool off too much while you swap them before taking new readings.

THE OUTPUT TRANSFORMER SHUNT METHOD

This is the way many pro techs measure plate current. A **good** quality DMM is required for this measurement. (When it comes to good DMMs, you have three choices...Fluke, Fluke, and Fluke.) This section assumes that you know a bit more about your amp, and how to use your testgear. If any of it is unclear, DON'T TRY THIS.

NOTE... Marshall amps have output transformers which have a very low DC resistance in the primary winding. If your meter's internal current-measuring shunt resistor is a relatively high value (~ 10 ohms, for instance) it will induce significant error into a transformer shunt measurement. This is because when such a meter is connected in parallel with half of the output transformer's primary, a significant portion of the current is not flowing through the meter, and can't be read. For this reason, unless you're **sure** you have a meter with a low internal current-sensing resistor (~ 1 ohm) the shunt method is **not** recommended for use on Marshall (and other low DCR) output transformers. Fairly good results can be achieved on Fenders, though.

WHAT YOU WILL DO

A)Read the current flowing through each leg of the output transformer's primary.B)Read the plate voltage.C)Use the above readings to calculate the static dissipation wattage.D)Adjust the bias to obtain the best tone, while keeping the tubes within specifications.

For this particular reading, you'll need to change your test leads to the CURRENT input jacks, and select the 200 mA DC range. The two probes are applied to the center tap and either of the ends of the output transformer's primary. (On a Fender, for instance, the center-tap is RED, and the two plate wires are BLUE and BROWN. On a Marshall, the center tap is usually BROWN, and the plate leads are usually RED and WHITE.)

On some amplifiers, the easiest way is to put one probe on pin THREE of either socket (or of either of the two sockets on each side) and the other on the center-tap, which will be located at some distance from the socket. Some amps (like the Marshall JCM 900 series, for instance) have all the wires soldered to terminals on the bottom of the output transformer, conveniently sticking up right where you can reach them.

The current that would normally flow through half of the transformer's primary winding is "shunted" through the meter, and thus measured. A small amount still flows through the part of the winding you are shunting, but the transformer's resistance is much higher than your meter's internal resistance. (See "NOTE" above.) Nearly all of the current flows through the meter.

SAFETY ADVISORY

BE WARNED...for all practical purposes, a meter set to measure CURRENT is equivalent to a STRAIGHT WIRE. This means that as soon as you touch either probe to the high voltage circuitry, THE OTHER PROBE NOW CARRIES THE SAME VOLTAGE. If you drop the probe and it lands on your arm or leg, you could be electrocuted. If it lands on the chassis (or anything else that is at earth or circuit ground potential) a huge spark will be generated, along with a noise like a small firecracker. (Please don't ask how I know this. ;-)

The probe tip will be partially melted, and at the very least, the meter's internal fuses will blow. At worst, the meter will be history. Shorting the HV to ground isn't especially good for the amp either, and may blow the amp's fuse or damage the circuitry. You can easily kill a rectifier tube this way. BE ESPECIALLY CAREFUL NOT TO LET A PROBE SLIP OFF A TERMINAL AND HIT THE CHASSIS WHILE YOU ARE TAKING A READING! BE *EXTRA* CAREFUL TO MAKE SURE YOUR FINGERS DON'T SLIDE DOWN THE PROBE AND COME INTO CONTACT WITH THE METAL TIP!! And make DOUBLE DAMN SURE you know which two points in the circuit you are supposed to touch the probes to, because if you accidentally touch the bias supply and the plate supply at the same time, you won't *believe* what happens. IF YOU'RE NOT *SURE* WHAT TO PROBE, *DON'T* PROBE IT!!

Once you've obtained the current readings from both sides of the output transformer's primary, you'll need to take a plate voltage reading so you can calculate the static dissipation wattage (as outlined above in the CATHODE RESISTOR method) and decide whether you need to increase or decrease the plate current. Note that if you are using the OPT shunt method with an amplifier which uses more than one tube per side on the transformer, you will need to divide the current reading on each side by the number of tubes used. Example: you read 88 mA on one side of a Twin Reverb's output tranny; that's 44 mA per tube, since there are two on each side. (4 total.)

REMEMBER TO REMOVE THE TEST LEADS FROM THE CURRENT MEASURING JACKS, AND TO SET THE METER TO THE HIGHEST DC VOLTAGE RANGE BEFORE YOU TRY TO READ THE PLATE VOLTAGE!! If you attempt to read the plate voltage with your meter still set up for a current reading, the results will be spectacular (as outlined above.) Since you may need to take several plate CURRENT and several plate VOLTAGE readings before you are finished setting the bias, you will need to be extremely vigilant about changing the meter settings (and the test leads) each time you take the different readings. Most pro techs use TWO METERS for this procedure, leaving one set up for current and one for voltage. (I use a handheld meter for the voltage reading, and a bench meter for the current.)

Once you have the necessary readings, the procedure is the same as for the CATHODE RESISTOR method: read, multiply, listen, adjust, read, multiply, listen, adjust, read, multiply, etc. Don't neglect the "settling" period, either. BE CAREFUL!!

TYPES OF (FIXED) BIAS CIRCUITS

Many amps which use "fixed" (negative grid) bias have provisions for adjusting the negative grid voltage upward or downward. Making the grids LESS negative will cause MORE current to flow through the tubes. Some amplifiers don't have a bias-adjusting control (pot) but instead use a fixed resistor to set the voltage. If you encounter one with a fixed resistor, the best thing to do is convert it to an adjustable type. Most of the time, the fixed resistor will be in parallel with the bias capacitor; the lower this resistor's value is, the lower the bias voltage will be. If you can locate and identify this resistor, you can replace it with a simple network consisting of a (lower value) resistor in series with a potentiometer. What you'll be shooting for is a range of adjustment that goes from LESS voltage to MORE voltage than is set by the (existing) fixed resistor. Take the value of the fixed resistor and divide by two; pick the closest standard value to your result, and put it in series with a pot which is as close to the original resistor's value as you can find. Example: the existing resistor is 33K; use a 15K resistor in series with a 25K pot to replace it. The original resistor was 33K; you now have the ability to adjust the value from 15K to 40K. This should provide you with sufficient adjustment range to set any plate current you wish. If not, use a different value pot or resistor.

Some amps have a "balance" type bias adjustment, which allows you to vary the negative grid voltage between the two halves of the output stage; this makes a "matched" set of tubes less crucial to good performance, although it can't compensate for tubes that are wildly different. If you encounter this circuit, the easiest way to adjust it is to simply "tune" the control for minimum 120Hz hum on the output. This type can be modded to the *best* type, which is not only variable from side-to-side, but adjustable up-and-down, too. Usually, this circuit will have the "balance" pot's wiper connected to a resistor which is grounded at the other end. You can replace this resistor exactly as outlined above (half the value, add a pot, etc.) and have the best of both worlds.

If the simple mods outlined above (and the reasons for making them) don't seem perfectly clear to you, DON'T TRY THEM. A schematic (and the expertise with which to interpret it) will go a long way towards helping you do them correctly. You can have the mods performed by a tech, and then do your own biasing from then on, if you wish.

TABLE A

Suggested MAX static dissipation wattages for common guitar amplifier tubes. You can exceed these (although I wouldn't do it with a 6V6) at the cost of some tube lifetime. The colder you run 'em, the longer they will last. Remember, as long as you don't run the tubes hot enough to damage them, there are *no* rules about how much current to set them for. If you like the way your amp sounds when your 6L6s are only pulling 14 watts, bully for you... you probably won't need to retube it for 10 years. I know that many sources for biasing information just specify plate (or cathode) current settings; telling you to bias your 6L6s at "35 milliamps" is nonsense. Unless you take the plate voltage into consideration, a current specification is meaningless. For instance, 40 mA at 250 volts is 10 watts; the same 40 mA at 500 volts is 20 watts... TWICE as much. In both cases, the current is the same. Amps vary; two identical amps can have plate voltages which differ by as much as 20%. Just because you have a schematic that specifies the plate voltage in your amp as being at 450VDC, don't expect to see that voltage when you take a measurement. TAKE the reading, don't assume the voltage will be as specified. Trust your meter. Most of these suggested MAX wattages have been arrived at through my own experience. NOTE THAT THESE FIGURES ARE NOT TARGETS, BUT MAXIMUMS. This is important. You are not looking to set your static plate dissipation to the values listed here, but to set it at a level which produces the tone you are looking for without exceeding them.

6V6 12 watts MAX 6L6GC (and variants, like the 7581A) 23 watts MAX 5881 (American) 18 watts MAX 5881 (Russian) 24 watts MAX EL34 20 watts MAX KT66 24 watts MAX 6550 27 watts MAX

KT88, KT90, KT100 can be treated as 6550s, although all three of these tubes are supposed to be able to take more current. The ultimate test is to view the tubes' plates IN THE DARK, after they have been powered up for 15-20 minutes. If you see any red spots, back the current off a bit. One exception to this is the NOS 6V6; some of these will show a slight red "stripe" down the center of the plates even when they're set fairly cold. I've seen them run for years in this condition. *Large* red blotches, or even the entire plates turning red, is what you want to watch out for.

Lord Valve (LORD_VALVE@prodigy.net)

VISIT MY WEBSITE: <http://www.nebsnow.com/LordValve>"><http://www.nebsnow.com/LordValve>

I specialize in top quality HAND SELECTED NOS and current-production vacuum tubes for guitar and bass amps. Good prices, fast service.

QSC amps, RNC compressors, lots of other good stuff!

NBS Electronics, 230 South Broadway, Denver, CO 80209-1510
Phone orders/tech support after 1:30 PM Denver time at 303-778-1156

VISA - MASTERCARD

"I'm not an asshole, but I *play* one on the Internet." - Lord Valve

Entire contents copyright Lord Valve, 08/25/01